Viresh Hunasagi

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Professional Summary

Data Science Engineering student with hands-on experience in developing machine learning models for crop nutrient deficiency detection. Implemented CNN and Random Forest models achieving up to 93% accuracy. Strong foundation in Python, data preprocessing, and model evaluation. Seeking an internship to apply analytical and data-driven problem-solving skills.

Skills

Programming: Python, C, TypeScript

Machine Learning & Deep Learning: TensorFlow, Keras, CNN, LSTM, Random Forest, SVM, Scikit-learn, LLM / AI

Web & Application Frameworks: Next.js, React, Node.js

Data Analysis: Pandas, NumPy, Matplotlib

Data Visualization: Tableau, Power BI

Databases & Tools: Git, GitHub, Docker, SerpAPI, PhonePe API

Core Concepts: Data Structures, OOP (Python), DBMS, Operating Systems, Computer Networks

AI & Image/Signal Processing: OpenCV, Librosa, Image Processing, Feature Engineering, MFCC

Projects

1. **Detection of Nutrient Deficiency in Crop Leaves**
   Python, CNN, Random Forest, OpenCV, Scikit-Learn, TensorFlow

   - Designed an AI-based system to detect Nitrogen, Phosphorus, and Potassium deficiencies from rice leaf images.
   - Implemented image preprocessing, segmentation, and feature extraction using RGB color features and GLCM-based texture analysis.
   - Developed Random Forest and CNN (MobileNetV2) models achieving up to 93% accuracy with strong real-world performance.

2. **WhatsAppForBusiness – Business Intelligence & Contact Discovery**
   Next.js, React, TypeScript, Node.js, SerpAPI, LLM/AI, Docker, PhonePe API

   - Developed a platform for discovering publicly available person and business information using name, email, phone number, and location-based searches.
   - Implemented public web search using SerpAPI with AI/LLM-based data extraction, validation, and duplicate removal.
   - Used Docker for application environment setup and integrated PhonePe API for payment processing.

3. **Acoustic Signal Processing for Sound Event Detection and 3D Localization (Ongoing)**

    - Working on detection and localization of sound events using signal processing and machine learning techniques.
    - Designing system pipeline and exploring feature extraction and spatial audio modeling approaches.

Education

Bachelor of Engineering: CSE (Data Science), Expected 2027
BLDEA's V P Dr PG Halakatti College of Engineering and Technology – Vijayapura
CGPA: 8.3

Pre-University Course (PCMB) – 91%
High School – 90%

Certifications

- Participated in Ideathon 2025 – Team-based prototype development
- NPTEL – Data Science Using Python – 2025
- NPTEL – Foundations of Deep Learning: Concepts and Applications (Elite + Silver) – 2026

Hobbies

- Staying updated through technology podcasts.
- Cooking and experimenting with new cuisines.